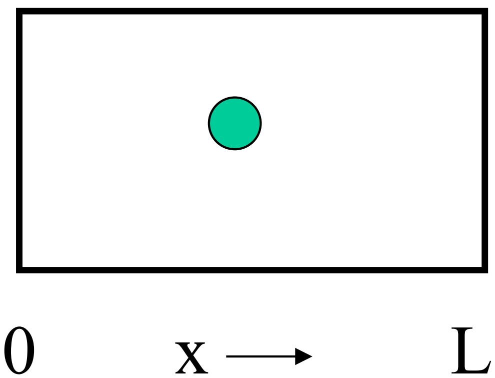
Particle in a Box

John W Daily

Particle in a Box

Motion is free except at walls so V(x) =0 except at walls, and



$$\psi(0) = \psi(L) = 0$$

$$\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = -\varepsilon_x \psi$$

Therefore $$\psi(x) = A\sin\sqrt{\frac{2m\varepsilon_x}{\hbar^2}}x + B\cos\sqrt{\frac{2m\varepsilon_x}{\hbar^2}}x$$

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1st BC $\longrightarrow$ $B = 0$

2nd BC $\longrightarrow$ $0 = A \sin \sqrt{\frac{2m\varepsilon_x}{\hbar^2}} L$

For this to be satisfied

$$\varepsilon_x = \frac{\hbar^2 \pi^2}{2mL} n_x^2$$

Where n_x is a positive integer

Thus, not all energies are allowed as stationary states of the particle, hence QUANTUM!

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Postulate 2 $\longrightarrow$ $\int_0^L |\psi|^2 \, dx = 1$ $\longrightarrow$ $A = \sqrt{\dfrac{2}{L}}$

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi n_x x}{L}\right) \qquad n_x = 0,1,2,\ldots$$

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In three dimensions

$$\varepsilon = \varepsilon_x + \varepsilon_y + \varepsilon_z = \frac{h^2}{8mV^{2/3}}(n_x^2 + n_y^2 + n_z^2)$$

$$\psi(x) = \sqrt{\frac{8}{L^3}} \sin\left(\frac{\pi n_x x}{L}\right) \sin\left(\frac{\pi n_y y}{L}\right) \sin\left(\frac{\pi n_z z}{L}\right)$$

Degeneracy

Note that different combinations of the three quantum numbers can result in the same energy

n_x	n_y	n_z	$n_x^2 + n_y^2 + n_z^2$	g
1	**1**	**1**	**3**	**1**
2	**1**	**1**	**6**	
1	**2**	**1**	**6**	**3**
1	**1**	**2**	**6**	

For large values of n, g can be very large

Some Numbers

$$m_{proton} = 1.67x10^{-27}\,kg \qquad h = 6.626x10^{-34}\,J\sec$$

$$k = 1.38x10^{-23}\,J/K$$

For a 1 cm^3 box $\qquad \dfrac{h^2}{8mV^{2/3}} = 3.28x10^{-37}\,J$

We shall see that $\qquad \langle \varepsilon \rangle = \dfrac{3}{2}kT$

So that

$$n_{typical} \sim \sqrt{n_x^2 + n_y^2 + n_z^2} = \left(\frac{8mV^{2/3}}{h^2} < \epsilon > \right)^{1/2} = 1.38 \times 10^8$$

Wave Functions